Activity 3.8 – CPU Bug Detection

1.Design Approach:

Designed a CPU bug detection support module using Verilog HDL to verify arithmetic addition, logical right shift and zero detection operations. Implemented combinational logic for checking the expected results.

2.Verification Steps:

- Created a separate testbench.
- Applied different input operands.
- Tested addition operations.
- Verified shift operations.
- Checked zero detection output.
- Compared actual and expected results.
- Simulated using ModelSim.

3.Results:

The testbench checks arithmetic, shift and zero detection outputs. PASS/FAIL messages are displayed.

4.Issues Faced:

Operand widths must be handled correctly. Shift amounts must be within the valid range. Zero detection must evaluate the complete result.

5.Conclusion:

Successfully designed and verified CPU arithmetic and logical checking operations using Verilog HDL.